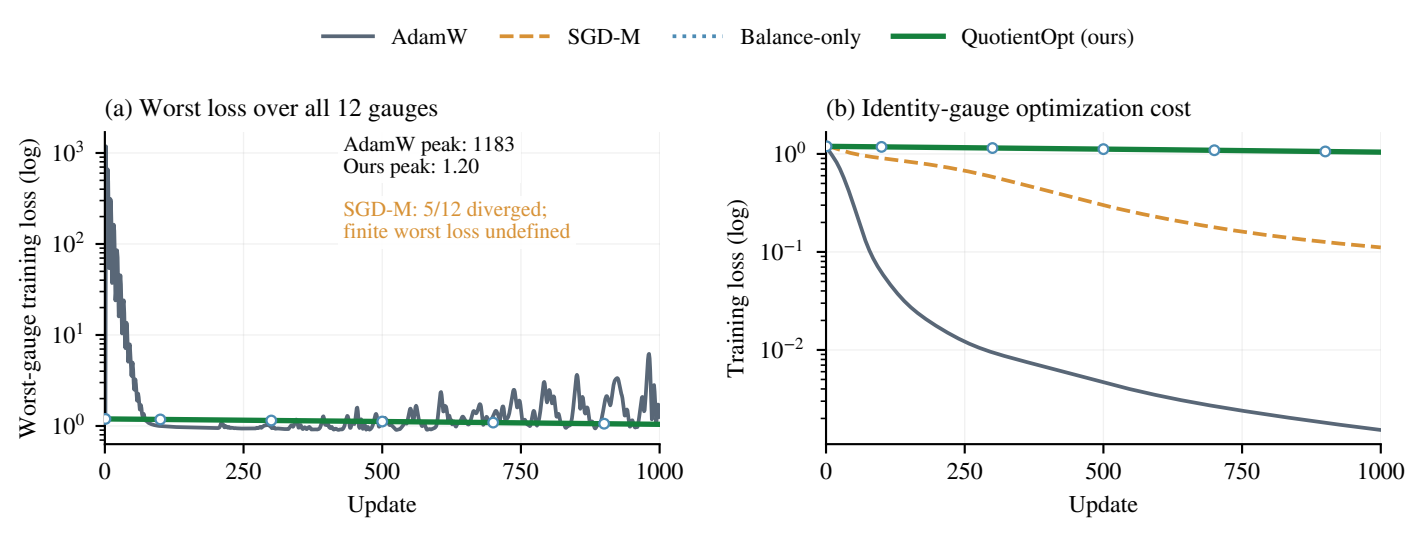
AdamW
SGD-M
Balance-only
QuotientOpt (ours)
(a) Worst loss over all 12 gauges
AdamW peak: 1183
Ours peak: 1.20
SGD-M: 5/12 diverged;
finite worst loss undefined
Worst-gauge training loss (log)
Update
(b) Identity-gauge optimization cost
Training loss (log)
Update